



Girraween High School

2016 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using a black or blue pen
- Board - approved calculators may be used
- A laminated reference sheet is provided
- Answer multiple choice questions on the front page
- In questions 11 – 16 start all questions on a separate page and show all relevant mathematical reasoning and/or calculations

Total Marks – 100

Section I Pages 5 - 8 10 marks

- Attempt 1 – 10
- Allow about 15 minutes for this section

Section II Pages 9 - 18 90 marks

- Attempt 11 – 16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt questions 1 – 10

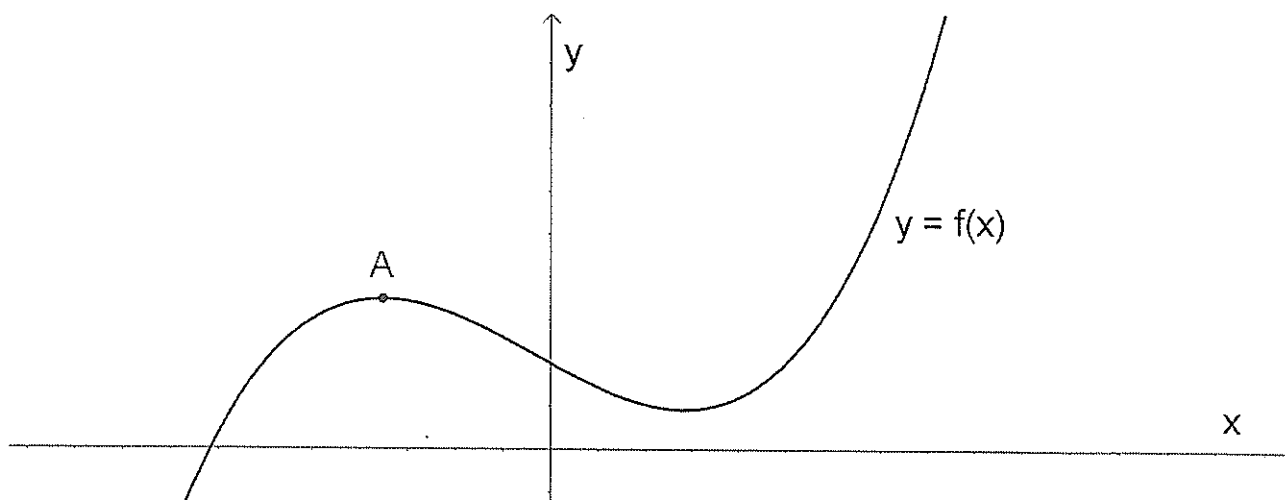
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

- 1 Find the value of $\log_e 2016$ to three significant figures.

- (A) 7.61
- (B) 7.60
- (C) 7.608
- (D) 7.609

- 2 The graph below shows the maximum stationary point A on the curve $y = f(x)$.



Which of the following is true at point A ?

- (A) $f'(x) > 0$ and $f''(x) = 0$
- (B) $f'(x) < 0$ and $f''(x) = 0$
- (C) $f'(x) = 0$ and $f''(x) > 0$
- (D) $f'(x) = 0$ and $f''(x) < 0$

- 3 The equation $2x^2 - 5x - 1 = 0$ has roots α and β .

What is the value of $\frac{1}{\alpha} + \frac{1}{\beta}$?

(A) $-\frac{1}{5}$

(B) 5

(C) -5

(D) $\frac{1}{5}$

- 4 The coordinates of the focus of the parabola $x^2 = 8(y - 3)$ are:

(A) (0, 5)

(B) (0, 1)

(C) (5, 0)

(D) (1, 0)

- 5 If $x = a(b - \frac{1}{y})$ then

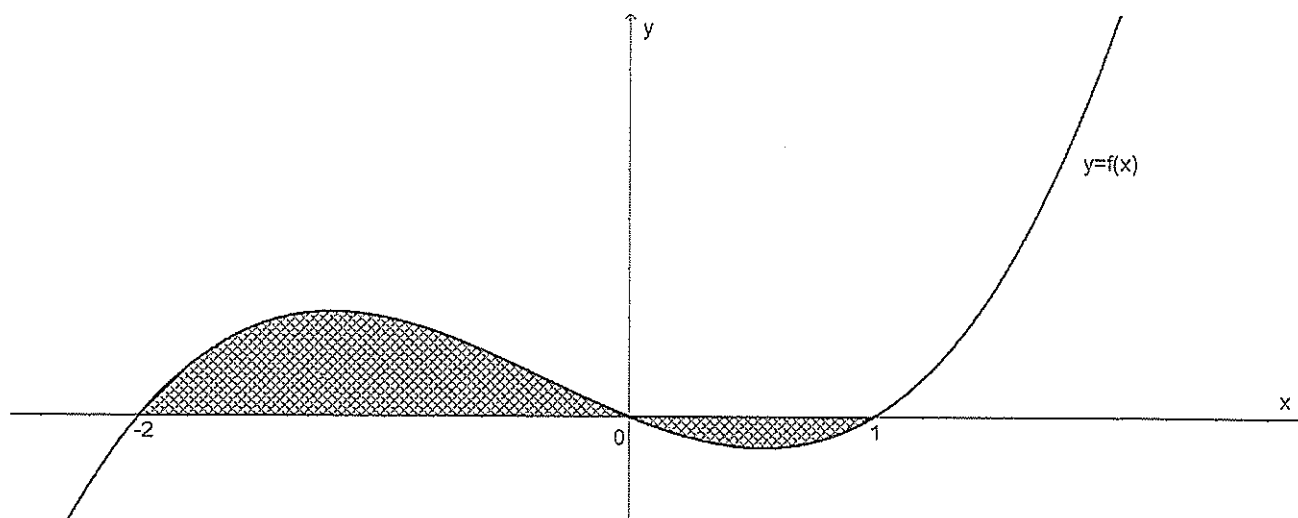
(A) $y = \frac{a}{b-x}$

(B) $y = \frac{a}{ab-x}$

(C) $y = \frac{1}{ab-x}$

(D) $y = \frac{x}{a} - b$

- 6 Which would give the value of the shaded area?



- (A) $\int_0^{-2} f(x)dx + \left| \int_0^1 f(x)dx \right|$
- (B) $\int_{-2}^0 f(x)dx + \left| \int_0^1 f(x)dx \right|$
- (C) $\left| \int_0^{-2} f(x)dx \right| + \int_0^1 f(x)dx$
- (D) $\left| \int_{-2}^0 f(x)dx \right| + \int_0^1 f(x)dx$
- 7 The solutions to $\sqrt{2}\sin x = -1$ for $0 \leq x \leq 2\pi$ are:

- (A) $\frac{3\pi}{4}$ and $\frac{5\pi}{4}$
- (B) $\frac{3\pi}{4}$ and $\frac{7\pi}{4}$
- (C) $\frac{5\pi}{4}$ and $\frac{7\pi}{4}$
- (D) $\frac{7\pi}{4}$ and $\frac{9\pi}{4}$

- 8 The solution to the inequality $6 - x - x^2 \leq 0$ is:
- (A) $-3 \leq x \leq 2$
 - (B) $x \leq -3$ or $x \geq 2$
 - (C) $x \leq -2$ or $x \geq 3$
 - (D) $-2 \leq x \leq 3$
- 9 The graph of $y = 3x^2 - kx + 2$ is symmetrical about the line $x = \frac{1}{2}$.
The lowest possible value of y is:
- (A) $\frac{1}{4}$
 - (B) $\frac{1}{2}$
 - (C) $\frac{3}{4}$
 - (D) $\frac{5}{4}$
- 10 What is the perpendicular distance between the lines $y = 4x + 3$ and $y = 4x + 5$?
- (A) $\frac{2}{\sqrt{17}}$
 - (B) $\frac{3}{\sqrt{17}}$
 - (C) $\frac{2}{5}$
 - (D) $\frac{3}{5}$

End of Section I

Section II

90 marks

Attempt questions 11 – 16

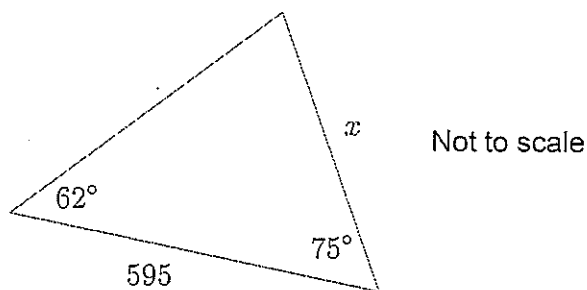
Allow about 2 hours 45 minutes for this section

Answer each section on a new page

In questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a new page.

- a) Simplify fully $3x - (4 - 3x)$. 1
- b) Factorise $x^3 + 8$ 1
- c) Solve $|2 - 5x| \leq 7$ 2
- d) Write $\frac{3-\sqrt{5}}{3+\sqrt{5}}$ in the form $a + b\sqrt{5}$, where a and b are rational. 2
- e) The side lengths of the triangle below are in millimetres.

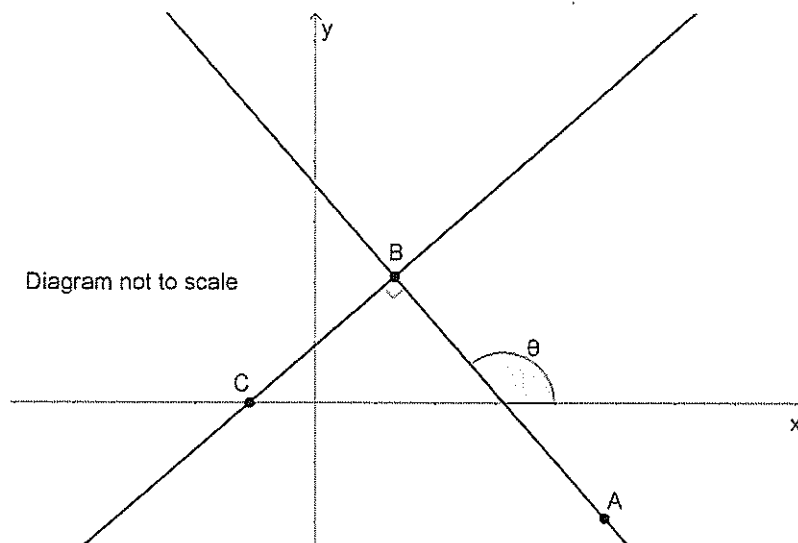


Find the value of x to the nearest whole number. 2

Question 11 continues on the next page

Question 11 continued

- f) The points $A(8, -3)$ and $B(5, 4)$ are shown in the diagram below. The line through AB makes an angle of θ with the positive x -axis and the point C lies on the x -axis.



- | | | |
|-------|--|---|
| (i) | Find the gradient of the line AB . | 1 |
| (ii) | Find the value of θ to the nearest degree. | 1 |
| (iii) | Find the coordinates of C given that $AB \perp BC$. | 2 |
| (iv) | Find coordinates of M , the midpoint of AB . | 1 |
| (v) | Find the equation of the line AB in general form. | 2 |

End of question 11

Question 12 (15 marks) Start a new page.

a) Find the gradient of the normal at the point $(2, -2)$ on the curve $y = x^3 - 5x$. 2

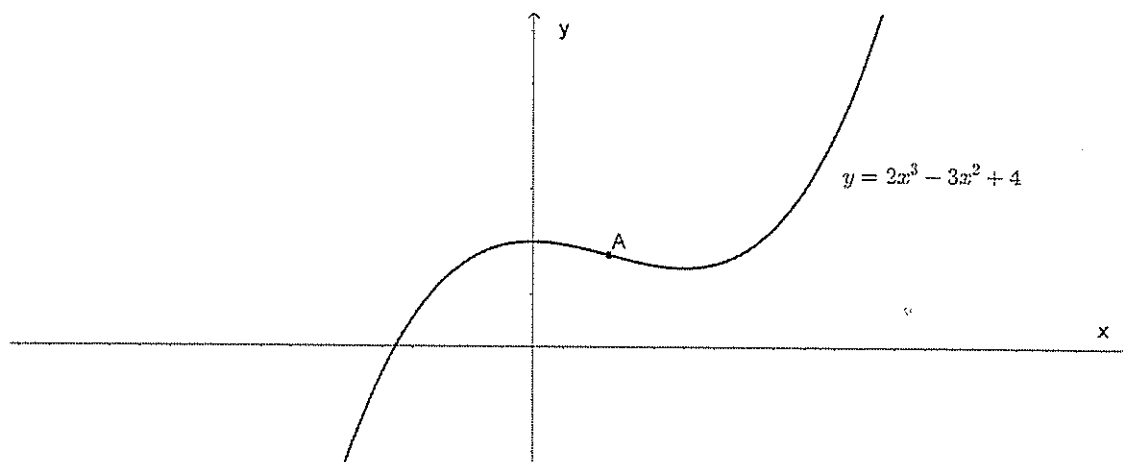
b) Differentiate with respect to x .

(i) $x \log_e(3x^2 - 1)$ 2

(ii) $(e^{-2x} + 1)^{10}$ 2

(iii) $\frac{5x}{\sin 2x}$ 2

c) The graph below shows the curve $y = 2x^3 - 3x^2 + 4$. The point A is a point of inflexion.



(i) Find the coordinates of A . 2

(ii) When is the curve concave up? 1

d) The number of bacteria (B) in a sample grows exponentially with time according to the equation $B = 200e^{kt}$, where k is a constant and t is measured in hours.

(i) In two days (48 hours) the number of bacteria in the sample is now 1653. Calculate the value of k to three decimal places. 2

(ii) Find, correct to the nearest hour, when there will be one million bacteria in the sample. 2

End of question 12

Question 13 (15 marks) Start a new page.

- a) The point $(-2, 3)$ lies on the curve with a gradient function of $\frac{dy}{dx} = \frac{4}{x+3}$.
Find the equation of the curve.

2

- b) Find the following integrals.

(i) $\int (6\cos 3x - 2\sin \frac{x}{2}) dx$

2

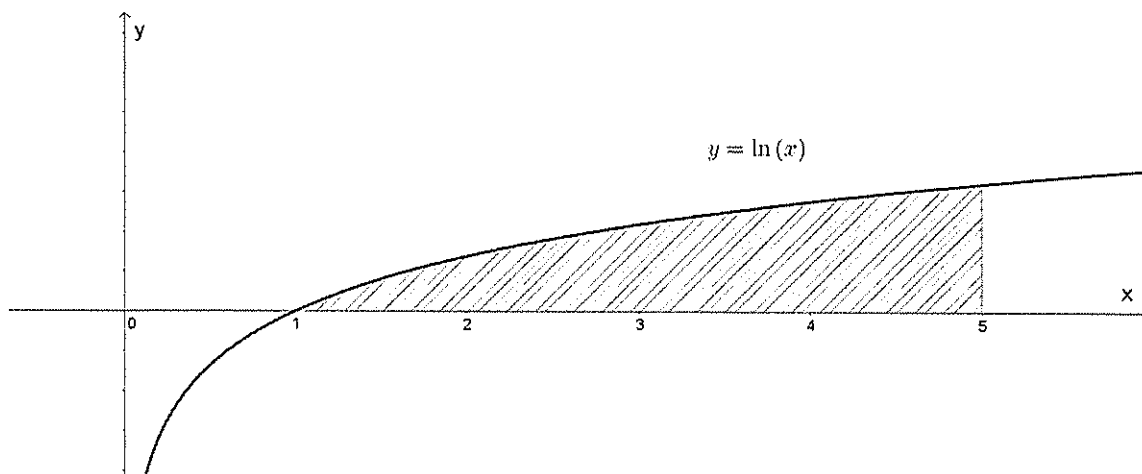
(ii) $\int \frac{6}{e^{3x}} dx$

2

(iii) $\int (1 - 6\sec^2 \frac{x}{3}) dx$

2

- c) The graph below shows the curve $y = \log_e x$.



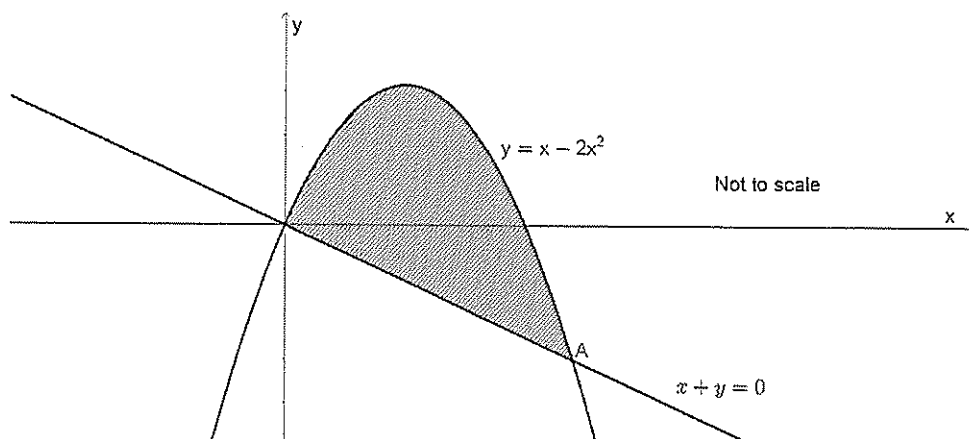
Use Simpson's Rule with five (5) function values to approximate $\int_1^5 \log_e x dx$. Give your answer to three (3) significant figures.

3

Question 13 continues on the next page

Question 13 continued

- d) The graph below shows the area enclosed by the parabola $y = x - 2x^2$ and the line $x + y = 0$. The parabola and the line intersect at the origin and point A .



- | | | |
|------|-------------------------------------|----------|
| (i) | Find the coordinates of point A . | 1 |
| (ii) | Find the value of the shaded area. | 3 |

End of question 13

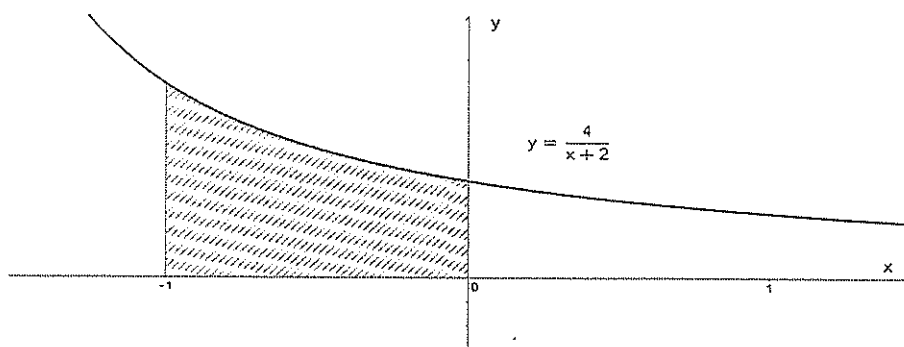
Question 14 (15 marks) Start a new page.

- a) An arithmetic sequence begins with the three terms $-6, 1, 8$.
- (i) Find the 200th term of the sequence. 2
 - (ii) Find the sum of 200 terms of the sequence. 2
- b) A geometric sequence begins with the three terms $-4, 8, -16$.
Find the 15th term of the progression. 2
- c) The numbers p, q and r add to 9 and form an arithmetic progression. The numbers r, p and q form a geometric progression. Find the values of p, q and r . 3
- d) On the 1st January each year Simone invests $\$M$ annually into a superannuation account. The account gives interest at a rate of 5% per annum, compounded annually.
- (i) Show that the value of her investment at the end of 2 years was $A_2 = 2.1525M$ dollars. 2
 - (ii) Show that the value of her investment at the end of n years was $A_n = 21(1.05^n - 1)M$ dollars. 2
 - (iii) Simone wants to retire after 30 years with a million dollars in her superannuation account. Find the amount that she must invest into her account on the 1st January each year to reach her goal. Answer to the nearest cent. 2

End of question 14

Question 15 (15 marks) Start a new page.

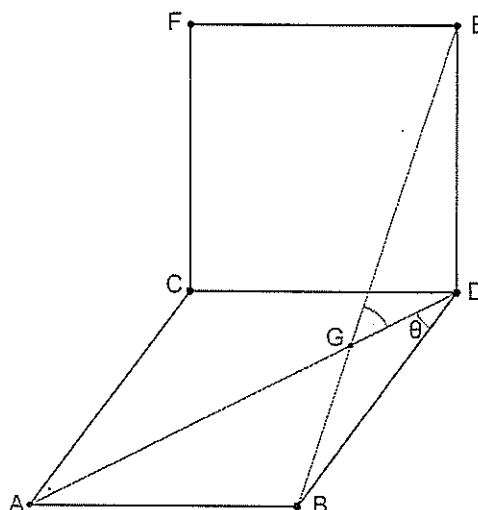
- a) The region bounded by the curve $y = \frac{4}{x+2}$, the line $x = -1$ and the axes is shown below.



The region is rotated about the x – axis to form a solid.
Find the volume of this solid.

3

- b) The diagram below shows square $CDEF$ and rhombus $ABDC$.
The diagonal of the rhombus AD and the segment BE intersect at point G .



- (i) Given that $\angle ADB = \theta$, explain why $\angle CDA = \theta$, giving reasons. 1
- (ii) Find $\angle BED$ in terms of θ , giving reasons. 2
- (iii) Hence show that $\angle DGE = \frac{\pi}{4}$, giving reasons. 2

Question 15 continues on the next page

- c) In a large country town, it is known that 55% of the population is male and 45% of the population is female. Three people in the town are surveyed at random. Find to the nearest percent, the probability that two are male and one is female. 2
- d) Albert plays a game where he throws two standard six-sided dice and the total of the faces showing is noted. Albert wins the game if an 8 is thrown and he loses if a 5 is thrown. If the sum is any other number, the game continues until an 8 is thrown or a 5 is obtained.
- (i) Show that the probability that Albert wins on the first throw is $\frac{5}{36}$. 1
- (ii) Show that the probability that Albert wins on either the first, second or third throw is $\frac{185}{576}$. 2
- (iii) What is the probability that Albert wins the game? 2

End of question 15

Question 16 (15 marks) Start a new page.

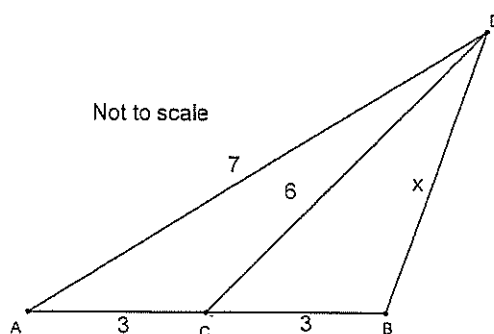
- a) Two particles P and Q which are initially at the origin are moving along a straight line. Their displacements, x kilometres, from the origin at any time, t hours, are given by the rules:

$$P: x = 5t - 2t^2.$$

$$Q: x = 8t^2 + 2t.$$

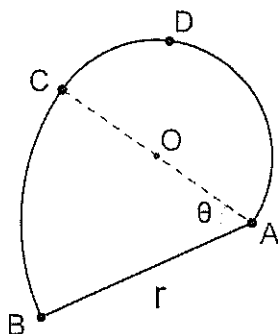
- (i) After what time are they travelling with the same velocity? 2
- (ii) Both particles are together again at point A . Find the distance of point A from the origin. 2
- (iii) A third particle R , travelling with constant speed, is 3 kilometres ahead of P and Q when they pass the origin. If particle R arrives at point A at the same time as particles P and Q , find a rule connecting x and t for this particle. 2
- b) Triangle ABD has side lengths of $AD = 7$ units, $DB = x$ units and $AB = 6$ units.

C is the midpoint of AB . The median CD equals the length of the base AB .



- (i) Use the cosine rule in triangle ADC to show that $\cos \angle DAB = \frac{11}{21}$. 1
- (ii) Hence find the exact value of x . 2

- c) The shape $ABCD$ consists of a sector ABC of radius r and angle θ and semicircle ACD with centre O and radius $\frac{r}{2}$.



- (i) If the area (A) of this shape is a fixed value, show that the perimeter $P = \left(\frac{\pi+4}{4}\right)r + \frac{2A}{r}$. 3
- (ii) Show that perimeter P is a minimum when $\theta = 1$. 3

End of examination

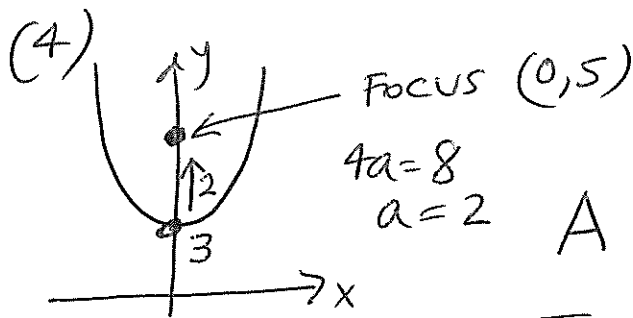
2016

TRIAL MATHEMATICS

(1) A

(2) D

$$(3) \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-b/a}{c/a} = \frac{5/2}{-1/2} = -5 \quad C$$



$$(5) x = a(b - \frac{1}{y})$$

$$\frac{x}{a} = b - \frac{1}{y}$$

$$\frac{1}{y} = b - \frac{x}{a} = \frac{ab - x}{a}$$

$$y = \frac{a}{ab - x} \quad B$$

(6) B

(7) $\sin x = -\frac{1}{\sqrt{2}}$

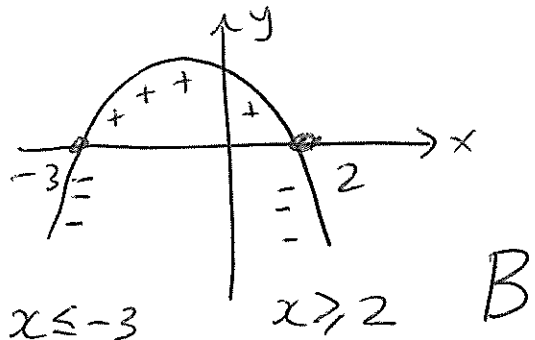
S	A
T	C

$$x = \pi + \frac{\pi}{4}, 2\pi - \frac{\pi}{4}$$

$$x = \frac{5\pi}{4}, \frac{7\pi}{4} \quad C$$

$$(8) 6 - x - x^2 \leq 0$$

$$(3 + x)(2 - x) \leq 0$$



(9) $\frac{dy}{dx} = 0$ at vertex when $x = \frac{1}{2}$

$$6x - k = 0$$

$$6x \frac{1}{2} - k = 0$$

$$k = 3$$

$$\therefore y = 3x^2 - 3x + 2$$

$$\therefore y = 3\left(\frac{1}{2}\right)^2 - 3\left(\frac{1}{2}\right) + 2 = 1\frac{1}{4} \quad D$$

(10) (0, 3) ON $y = 4x + 3$

DISTANCE TO $4x - y + 5 = 0$

is $\left| \frac{4 \times 0 - 3 + 5}{\sqrt{4^2 + (-1)^2}} \right| = \frac{2}{\sqrt{17}}$

A

QUESTION 11

$$(a) \quad 3x - 4 + 3x = 6x - 4$$

$$(b) \quad (x+2)(x^2 - 2x + 4)$$

$$(c) \quad 2-5x \leq 7, \quad -(2-5x) \leq 7$$
$$-5x \leq 5 \quad -2+5x \leq 7$$
$$x \geq -1 \quad x \leq 1.8$$
$$\text{I.e. } -1 \leq x \leq 1.8$$

$$(d) \quad \frac{3-\sqrt{5}}{3+\sqrt{5}} \times \frac{3-\sqrt{5}}{3-\sqrt{5}}$$

$$= \frac{9 - \cancel{6\sqrt{5}} + 5}{9 - 5}$$

$$= \frac{14 - \cancel{6\sqrt{5}}}{4}$$

$$= \frac{7}{2} - \frac{3}{2}\sqrt{5}$$

$$a = \frac{7}{2} \quad b = -\frac{3}{2}$$

$$(e) \quad \frac{x}{\sin 62^\circ} = \frac{595}{\sin 43^\circ}$$

$$x \doteq 770$$

$$(f)(i) \quad m_{AB} = \frac{-3-4}{8-5}$$

$$= -\frac{7}{3}$$

$$(ii) \quad \tan \theta = -\frac{7}{3}$$

$$\theta = 180^\circ - 67^\circ = 113^\circ$$

$$(iii) \quad \text{Let } C \text{ be } (c, 0)$$

$$m_{BC} \cdot m_{AB} = -1$$

$$\frac{0-4}{c-5} \cdot \frac{-7}{3} = -1$$

$$\dots \quad 28 = -3(c-5)$$

$$28 = -3c + 15$$

$$13 = -3c$$

$$c = -4\frac{1}{3}$$

$$\therefore C(-4\frac{1}{3}, 0)$$

$$(iv) \quad \left(\frac{8+5}{2}, \frac{-3+4}{2} \right) = \left(6\frac{1}{2}, \frac{1}{2} \right)$$

$$(v) \quad y - y_1 = m(x - x_1)$$

$$y - 4 = -\frac{7}{3}(x - 5)$$

$$3(y - 4) = -7(x - 5)$$

$$3y - 12 = -7x + 35$$

$$7x + 3y - 47 = 0$$

QUESTION 12

$$(a) y = x^3 - 5x$$

$$\frac{dy}{dx} = 3x^2 - 5$$

$$x=2 \quad \frac{dy}{dx} = 3(2)^2 - 5 = 7$$

$$m_1 = 7$$

$\therefore$ Gradient of normal

$$m_2 = -\frac{1}{m_1} = -\frac{1}{7}$$

(b) Product Rule

$$(i) x \frac{d}{dx}(\ln(3x^2-1)) + \ln(3x^2-1) \frac{d}{dx}(x)$$

$$= x \left(\frac{6x}{3x^2-1} \right) + \ln(3x^2-1) \cdot 1$$

$$= \frac{6x^2}{3x^2-1} + \ln(3x^2-1)$$

(ii) Chain Rule

$$10(e^{-2x} + 1)^9 \frac{d}{dx}(e^{-2x} + 1)$$

$$= 10(e^{-2x} + 1)^9 (-2e^{-2x})$$

$$= -20e^{-2x}(e^{-2x} + 1)^9$$

(iii) Quotient Rule

$$= \frac{\sin 2x \frac{d}{dx}(5x) - 5x \frac{d}{dx}(\sin 2x)}{(\sin 2x)^2}$$

$$= \frac{5\sin 2x - 10x \cos 2x}{\sin^2 2x}$$

$$(c) y = 2x^3 - 3x^2 + 4$$

$$(i) y' = 6x^2 - 6x$$

$$y'' = 12x - 6$$

Point of Inflection $y'' = 0$

$$12x - 6 = 0$$

$$x = \frac{1}{2}$$

$$y = 2\left(\frac{1}{2}\right)^3 - 3\left(\frac{1}{2}\right)^2 + 4 = 3\frac{1}{2}$$

$$\therefore A\left(\frac{1}{2}, 3\frac{1}{2}\right)$$

$$(ii) \text{ Concave up: } y'' > 0$$
$$x > \frac{1}{2}$$

$$(d) (i) t = 48 \quad B = 1653$$

$$1653 = 200 e^{48k}$$

$$e^{48k} = 8.265$$

$$48k = \ln(8.265)$$

$$k = \frac{\ln(8.265)}{48} = 0.044$$

$$(ii) \boxed{B = 200 e^{0.044t}}$$

$$1000000 = 200 e^{0.044t}$$

$$e^{0.044t} = 5000$$

$$t = \frac{\ln 5000}{0.044} \approx 194 \text{ hours}$$

(about 8 days)

QUESTION 13

$$(a) y = 4 \log_e(x+3) + C$$

$$\text{SUB } (-2, 3)$$

$$3 = 4 \times 0 + C$$

$$\therefore C = 3$$

EQUATION

$$y = 4 \log_e(x+3) + 3$$

(b)

$$(i) -2 \sin 3x - 4 \cos \frac{x}{2} + C$$

$$(ii) -2e^{-3x} + C \text{ or } \frac{-2}{e^{3x}} + C$$

$$(iii) x - 18 \tan \frac{x}{3} + C$$

$$(c) \int_1^5 \log_e x \, dx = \int_1^3 \log_e x \, dx + \int_3^5 \log_e x \, dx$$

$$= \frac{3-1}{6} (\log_e 1 + 4 \log_e 2 + \log_e 3)$$

$$+ \frac{5-3}{6} (\log_e 3 + 4 \log_e 4 + \log_e 5)$$

$$= 1.2904 + 2.7510$$

$$\hat{=} 4.04$$

Solve simultaneously

$$(d) \begin{cases} y = -x \\ y = x - 2x^2 \end{cases}$$

$$(i) \begin{aligned} -x &= x - 2x^2 \\ 0 &= 2x - 2x^2 \\ 0 &= 2x(1-x) \end{aligned}$$

$$\begin{aligned} x=0 &\} & x=1 &\} \\ y=0 &\} & y=-1 &\} \end{aligned}$$

$$\text{So } A = (1, -1)$$

$$(ii) \text{Area} = \int (\text{top graph} - \text{bottom graph})$$

$$= \int_0^1 (x - 2x^2) - (-x) \, dx$$

$$= \int_0^1 2x - 2x^2 \, dx$$

$$= \left[x^2 - \frac{2}{3} x^3 \right]_0^1$$

$$= \left(1 - \frac{2}{3} \right) - 0$$

$$= \frac{1}{3} \text{ units}^2$$

QUESTION 14

(a) $a = -6$ $d = 7$

(a) $a = -6$ $d = 7$

$$T_{200} = -6 + 199 \times 7$$

$$T_{200} = 1387$$

$$(11) S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_{200} = \frac{200}{2} [-12 + 199 \times 7]$$

$$S_{200} = 138100$$

$$\textcircled{b} \quad T_n = ar^{n-1}$$

$$T_{15} = -4(-2)^{14}$$

$$T_{15} = -65536$$

(c) p, q, r form an AP

$$\therefore 2 - p = 7 - 2$$

$$2q = p + r$$

Also $q + p + r = 9$

$$2 + 29 = 9$$

$$3^2 = 9$$

$$9 = 3 \dots \textcircled{1}$$

and $p+r=6$ --- (2)

r, p, q form a GP

$$\frac{p}{r} = \frac{2}{p}$$

$$p^2 = r^2$$

Using ① & ②

$$p^2 = (6-p)3$$

$$p^2 = 18 - 3p$$

$$p^2 + 3p - 18 = 0$$

$$(p+6)(p-3)=0$$

$$\int P = -6 \quad \int P = 3$$

$$\begin{cases} \dot{q} = 3 \\ q = 3 \end{cases}$$

$$L^{\vee}_{r=12} \quad L^{\vee}_{r=3}$$

(d) $A_0 = M$

(1) $A_1 = M \times 1.05$

$$A_2 = (M \times 1.05 + M) \times 1.05$$

$\uparrow$ $\uparrow$
 NEXT INTEREST
 INVESTMENT

$$A_2 = M \times 1.05^2 + M \times 1.05$$

$$A_2 = (1.05^2 + 1.05)M$$

$$A_2 = 2.1525 \text{ M}$$

$$(11) A_3 = (A_2 + \underset{\uparrow}{M}) \times \underset{\uparrow}{1.05}$$

NEXT INTEREST
INVESTMENT

$$= (M \times 1.05^2 + M \times 1.05 + M) \times 1.05$$

$$= M \times 1.05^3 + M \times 1.05^2 + M \times 1.05$$

$$= (1.05^3 + 1.05^2 + 1.05)M$$

Continuing the pattern

$$A_n = (1.05^n + 1.05^{n-1} + \dots + 1.05) M$$

GP $\uparrow$ n terms
 $a = 1.05$
 $r = 1.05$

$$A_n = \frac{a(r^n - 1)}{r - 1} M$$

$$= \frac{1.05(1.05^n - 1)}{1.05 - 1} M$$

$$= \frac{1.05}{0.05} (1.05^n - 1) M$$

$$= 21(1.05^n - 1) M$$

(III) $A_n = 1000000 \quad n = 30$

$$1000000 = 21(1.05^{30} - 1) M$$

$$M = \frac{1000000}{21(1.05^{30} - 1)}$$

$$M = 14334.70$$

So each investment needs to be

$$\$14334.70$$

QUESTION 15

15(a) $V = \pi \int y^2 dx$

$$= \pi \int_{-1}^0 \frac{16}{(x+2)^2} dx$$

$$= 16\pi \int_{-1}^0 (x+2)^{-2} dx$$

$$= 16\pi \left[-(x+2)^{-1} \right]_{-1}^0$$

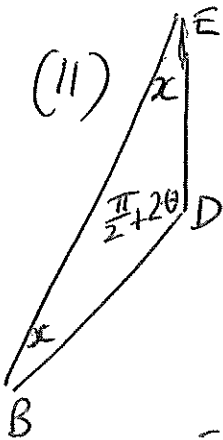
$$= 16\pi \left[\frac{-1}{x+2} \right]_{-1}^0$$

$$= 16\pi \left[\left(-\frac{1}{2}\right) - (-1) \right]$$

$$= 8\pi \text{ units}^3$$

(b)(i) $\angle CDA = \angle ADB = \theta$

(The diagonal of rhombus bisects the vertex angle)

(ii)  $\angle EDC = \frac{\pi}{2}$ (L in a square)

$$\therefore \angle EDB = \frac{\pi}{2} + 2\theta$$

$$DE = CD \text{ (sides of square)}$$

$$BD = CD \text{ (sides of rhombus)}$$

$$\therefore DE = BD$$

$$\therefore \angle BED = \angle EDB \text{ (L's opposite equal sides in } \triangle BDE)$$

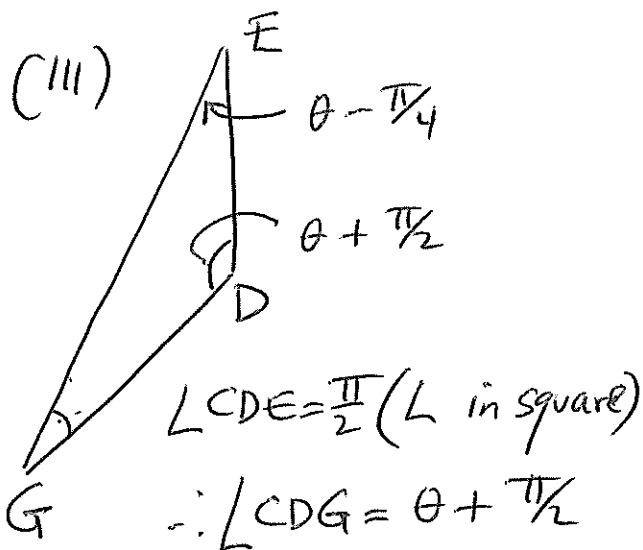
$$x + x + \left(\frac{\pi}{2} + 2\theta\right) = \pi$$

(angle sum of $\triangle$)

$$2x + 2\theta = \frac{\pi}{2}$$

$$x + \theta = \frac{\pi}{4}$$

$$\therefore \angle BED = x = \frac{\pi}{4} - \theta$$



$\angle DGE + \left(\frac{\pi}{4} - \theta\right) + \left(\theta + \frac{\pi}{2}\right) = \pi$
 (L sum of $\triangle DGE$)

$\angle DGE + \frac{3\pi}{4} = \pi$
 $\angle DGE = \frac{\pi}{4}$

(b) $P(MMF) + P(MFM) + P(FMM)$
 $= 3 \times .55 \times .55 \times .45$
 $\approx 41\%$

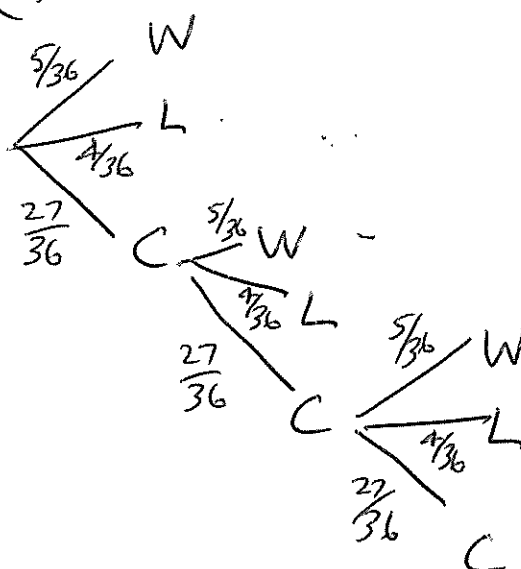
(c)

(i)

	1	2	3	4	5	6
1	2	3	4	(5)	6	7
2	3	4	(5)	6	7	(8)
3	4	(5)	6	7	(8)	9
4	(5)	6	7	(8)	9	10
5	6	7	(8)	9	10	11
6	7	(8)	9	10	11	12

$P(8) = \frac{5}{36}$

(ii) $W = \text{WIN}$ $L = \text{LOSE}$ $C = \text{CONTINUE}$



$P = \frac{5}{36} + \frac{3}{4} \times \frac{5}{36} + \frac{3}{4} \times \frac{3}{4} \times \frac{5}{36}$
 $= \frac{5}{36} \left(1 + \frac{3}{4} + \left(\frac{3}{4}\right)^2\right)$
 $= 185/576$

(iii) Continuing pattern above indefinitely

$P = \frac{5}{36} \left(1 + \frac{3}{4} + \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^3 + \dots\right)$
 LIMITING SUM

$= \frac{5}{36} \left(\frac{a}{1-r}\right)$

$= \frac{5}{36} \left(\frac{1}{1-\frac{3}{4}}\right)$

$= \frac{5}{9}$

QUESTION 16

(a) P: $V = 5 - 4t$

(i) Q: $V = 16t + 2$

$$5 - 4t = 16t + 2$$

$$t = \frac{3}{20} \text{ hour (9 mins)}$$

(ii) $5t - 2t^2 = 8t^2 + 2t$

$$10t^2 - 3t = 0$$

$$t(10t - 3) = 0$$

$$t = 0 \quad t = \frac{3}{10} \text{ hour}$$

(at origin) (at A)

$$\therefore x = 5\left(\frac{3}{10}\right) - 2\left(\frac{3}{10}\right)^2 = 1.32$$

So A is 1.32 km from origin

(iii) Particle R $v = c_1$
 $x = c_1 t + c_2$

$$t = 0 \quad x = 3 \quad \therefore c_2 = 3$$

$$t = \frac{3}{10} \quad x = 1.32 \quad 1.32 = c_1\left(\frac{3}{10}\right) + 3$$

$$c_1 = -5.6$$

$$\therefore x = -5.6t + 3$$

(b) (i) $\cos \angle DAB = \frac{3^2 + 7^2 - 6^2}{2 \times 3 \times 7} = \frac{11}{21}$

(ii) $x^2 = 7^2 + 6^2 - 2 \times 6 \times 7 \times \cos \angle DAB$

$$x^2 = 7^2 + 6^2 - 2 \times 6 \times 7 \times \frac{11}{21}$$

$$x = \sqrt{41}$$

(c) (i) $A = \text{sector} + \text{semicircle}$

$$A = \frac{1}{2} r^2 \theta + \frac{1}{2} \pi \left(\frac{1}{2} r\right)^2$$

$$A = \frac{1}{2} r^2 \theta + \frac{1}{8} \pi r^2$$

$$8A = 4r^2 \theta + \pi r^2$$

$$4r^2 \theta = 8A - \pi r^2 \quad (\div 4r^2)$$

$$\theta = \frac{2A}{r^2} - \frac{\pi}{4} \quad \dots \dots *$$

$P = \text{radius} + \text{arc} + \text{semicircle}$

$$P = r + r\theta + \frac{1}{2} \times 2\pi \left(\frac{1}{2} r\right)$$

$$P = r + r\left(\frac{2A}{r^2} - \frac{\pi}{4}\right) + \frac{\pi r}{2}$$

$$P = r + \frac{2A}{r} - \frac{\pi r}{4} + \frac{\pi r}{2}$$

$$P = \frac{2A}{r} + r\left(1 + \frac{\pi}{4}\right)$$

$$P = \frac{2A}{r} + \left(\frac{4+\pi}{4}\right) r$$

(ii) $\frac{dP}{dr} = -\frac{2A}{r^2} + \left(\frac{4+\pi}{4}\right)$

$$\frac{d^2P}{dr^2} = \frac{4A}{r^3} > 0$$

$\therefore \frac{dP}{dr} = 0$ GIVES MINIMUM

$$\frac{2A}{r^2} = \frac{4+\pi}{4}$$

$$r = \sqrt{\frac{8A}{4+\pi}} \quad \text{since } r > 0$$

$\therefore \theta = \frac{2A}{\frac{8A}{4+\pi}} - \frac{\pi}{4} \quad \text{from } *$

$$\theta = \frac{4+\pi}{4} - \frac{\pi}{4} = 1$$

(radian)